PROJECT REPORT

NORTHWIND TRADE

Sales, Product & Customer Analysis using Microsoft Power BI

Submitted by

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Course: B.Com FinTech with AI

October 2026

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1. Introduction

Northwind Trade is a well-known sample trading company that imports and exports specialty food products to customers across the world. Its database records the complete trading cycle: the products on offer, the customers who buy them, the employees who take the orders, the shippers who deliver them and the value of every transaction.

This report presents a business intelligence study of the Northwind Trade data using Microsoft Power BI. The raw tables were loaded into Power BI, modelled, and converted into a four-page interactive dashboard: Sales Overview, Product & Customer Analysis, Sales Dashboard and Tool Tip. The report explains each page, shows the supporting charts and tables, and interprets what the numbers mean for the business.

2. Objectives of the Study

- To import, clean and model the Northwind Trade data in Power BI.
- To calculate key performance indicators (KPIs) such as total sales, total orders, total quantity, total freight and average freight.
- To analyse sales performance by month, product category and employee.
- To study the geographic spread of customers and identify the highest-selling products.
- To build an interactive dashboard with slicers and tool tips so decision-makers can explore the data themselves.
- To draw conclusions and recommendations that can support better trade decisions.

3. About Northwind Trade and the Data Model

The dataset is organised as a relational model of seven related tables, visible in the Data pane of Power BI. Each table stores one aspect of the business and is linked to the others through key columns (for example, orders are linked to customers, employees and shippers, and order details are linked to products).

Table	Description
categories	Product groups such as Beverages, Dairy Products, Confections, Seafood, Condiments, Produce, Meat & Poultry and Grains & Cereals.
customers	Customer companies with their country and company name; 91 customers in total.
employees	Sales staff who handled the orders (e.g., Margaret Peacock, Janet Leverling, Nancy Davolio).
orders	Order header information: order date, customer, employee, shipper and freight cost; 809 orders.
order_details	Line items of every order: product, unit price and quantity sold.

Table	Description
products	Product master list with product name and its category.
shippers	Logistics companies used to deliver the orders.

Table 1: Tables used in the Northwind data model

4. Tools and Methodology

The analysis followed a standard business-intelligence workflow in Power BI Desktop:

- **Get Data:** the Northwind tables were imported into Power BI.
- **Transform Data:** data types, column names and relationships were checked and cleaned in Power Query.
- **Modelling:** relationships were created between the tables and DAX measures were written for Total Sales, Total Orders, Total Quantity Sold, Total Freight, Average Freight and Number of Customers.
- **Visualisation:** KPI cards, line, column and bar charts, a table, slicers and a custom tool-tip page were designed with a consistent blue-and-green colour theme.
- **Interaction:** slicers (company name, order date and others) and cross-filtering were enabled so that every visual responds to the user's selections.

5. Page 1 – Sales Overview

The first page gives a quick summary of the company's performance. Six KPI cards at the top present the headline numbers, followed by three charts that break the performance down by time, product category and employee.

5.1 Key Performance Indicators

KPI	Value	Meaning
Total Sales	$1.27 M ($1,265,793.04)	Revenue from all orders
Total Orders	809	Number of orders placed
Number of Customers	91	Distinct customers who ordered
Total Quantity Sold	51K (51,317 units)	Units sold across all products
Total Freight	63.96 K	Shipping cost of all orders
Average Freight	79.05	Average freight per order

Table 2: Headline KPIs from the Sales Overview page

5.2 Additional Derived Metrics

Using the headline KPIs, the following ratios give deeper insight into the business:

Metric	Value	Calculation
Average sales per order	$1,564.64	Total Sales ÷ Total Orders
Average quantity per order	63.4 units	Total Quantity ÷ Total Orders
Average sales per customer	$13,909.81	Total Sales ÷ Customers
Average orders per customer	8.9	Total Orders ÷ Customers
Average revenue per unit	$24.67	Total Sales ÷ Total Quantity
Freight as % of sales	5.1%	Total Freight ÷ Total Sales

Table 3: Derived business ratios

5.3 Total Orders by Month

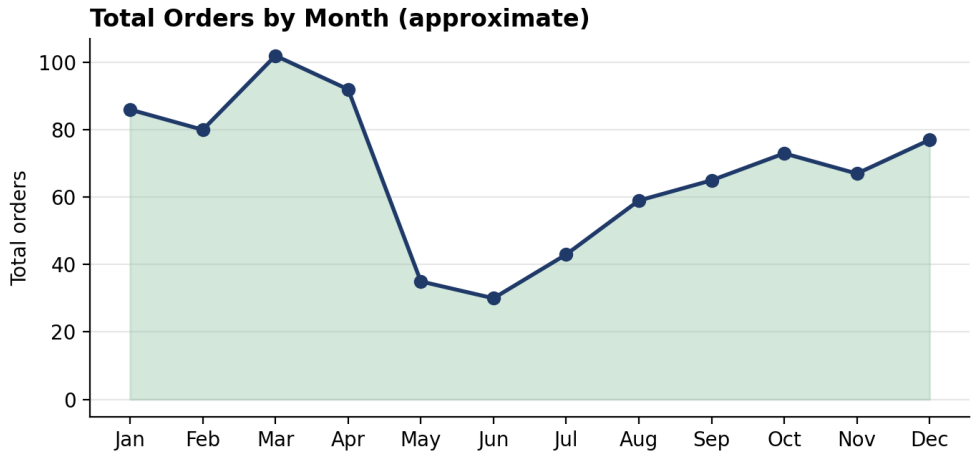


Figure 1: Total orders by month (values read approximately from the dashboard line chart)

Order volume is strongest at the start of the year, peaking around March at roughly 100 orders. It then falls sharply in May and June, which are the weakest months, before gradually recovering through the second half of the year. This indicates a clear seasonal pattern that should be considered in inventory and staffing plans.

5.4 Total Sales by Category

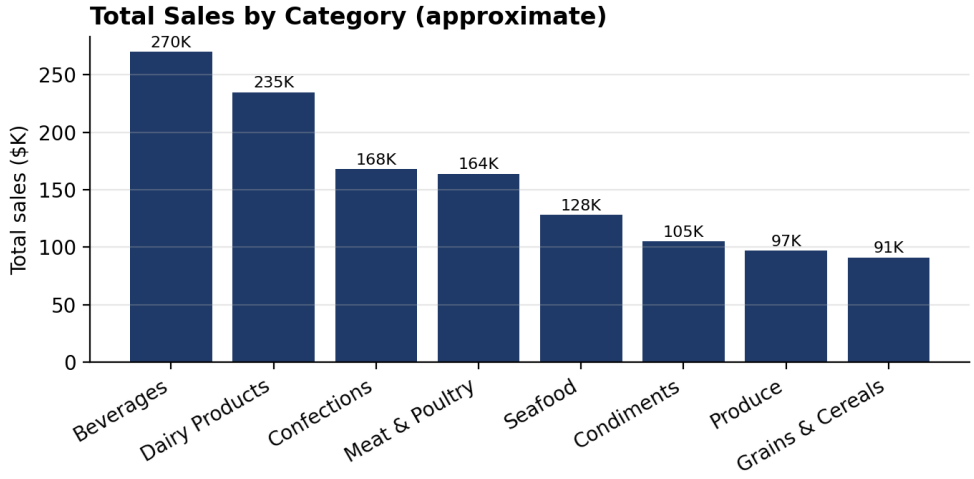


Figure 2: Total sales by product category (approximate values read from the dashboard chart)

Beverages is the leading category, followed by Dairy Products, together contributing roughly 40% of total sales. Confections and Meat & Poultry form the middle tier, while Seafood, Condiments, Produce and Grains & Cereals contribute the smallest shares.

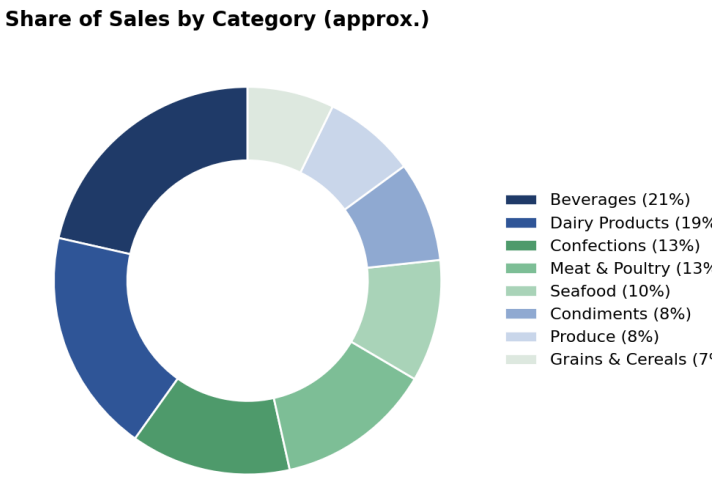


Figure 3: Share of sales by category (approximate)

5.5 Total Orders by Employee

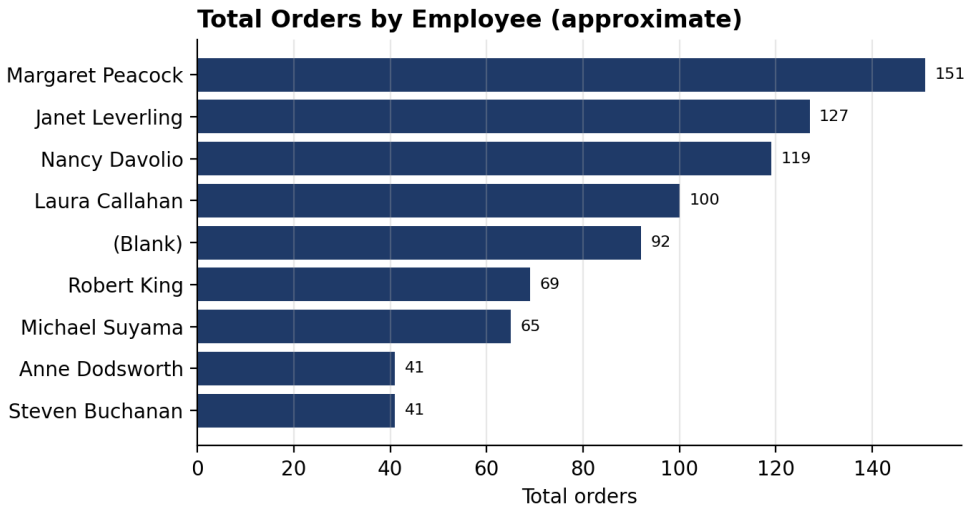


Figure 4: Total orders handled by each employee (approximate values read from the dashboard chart)

Margaret Peacock handled the most orders (151, about 19% of all 809 orders), followed by Janet Leverling and Nancy Davolio. Steven Buchanan and Anne Dodsworth handled the fewest. A "(Blank)" group also appears, meaning some orders have no employee assigned in the data, which is a data-quality point worth correcting.

6. Page 2 – Product & Customer Analysis

The second page focuses on who the customers are and which products drive revenue. It contains a bar chart of customers by country and a detailed product table showing category, total sales and total quantity sold.

6.1 Number of Customers by Country

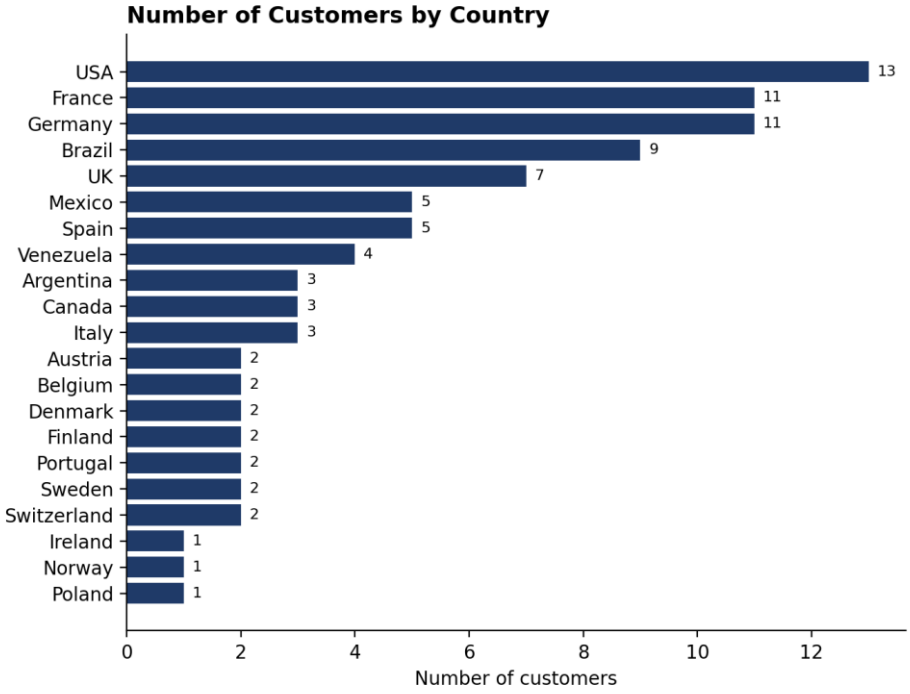


Figure 5: Number of customers by country

Country group	Countries	Customers	Share of customers
Top 3 (USA, France, Germany)	3	35	38%
Next tier (Brazil, UK, Mexico, Spain)	4	26	29%
Mid tier (Venezuela, Argentina, Canada, Italy)	4	13	14%
Small markets (Austria to Poland)	10	17	19%
Total	21	91	100%

Table 4: Customer concentration by country (customer counts read from the bar chart)

The United States has the largest customer base (about 13 customers), followed by France and Germany with about 11 each. These three countries alone account for nearly two-fifths of all customers, while countries such as Ireland, Norway and Poland have only one customer each. Customers are therefore concentrated in North America and Western Europe, with room to grow elsewhere.

6.2 Product Sales Table

The product table lists every product with its category, total sales and total quantity sold. The grand total row confirms $1,265,793.04 in sales and 51,317 units sold. The table below highlights the highest-selling products among those visible on the dashboard page.

#	Product	Category	Total Sales ($)	Quantity Sold
1	Côte de Blaye	Beverages	141,396.74	623
2	Camembert Pierrot	Dairy Products	46,825.48	1,577
3	Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1,263
4	Alice Mutton	Meat & Poultry	32,698.38	978
5	Carnarvon Tigers	Seafood	29,171.88	539
6	Ipoh Coffee	Beverages	23,526.70	580
7	Gudbrandsdalsost	Dairy Products	21,942.36	714
8	Ikura	Seafood	20,867.34	742
9	Gumbär Gummibärchen	Confections	19,849.14	753
10	Flotemysost	Dairy Products	19,551.03	1,057

Table 5: Highest-selling products visible in the dashboard table

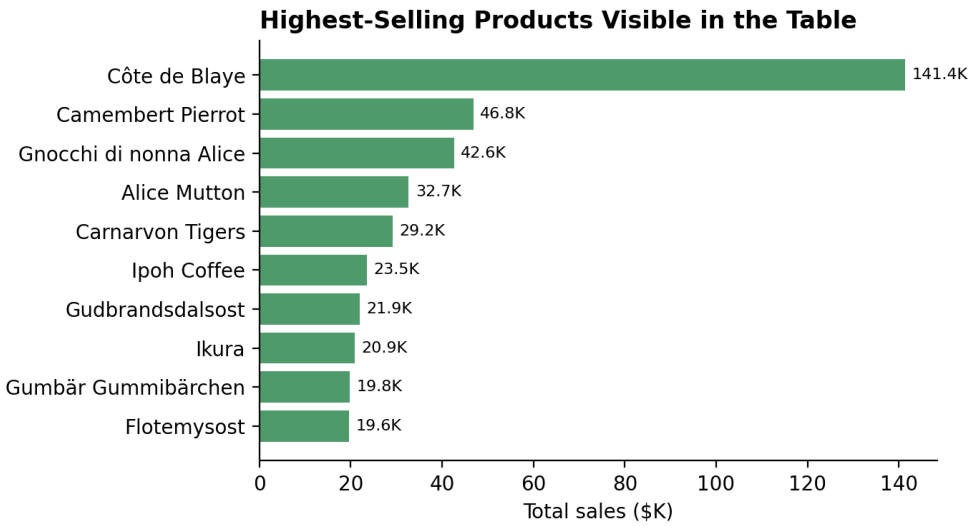


Figure 6: Highest-selling products (from the dashboard table)

Côte de Blaye is a clear outlier: with $141,396.74 in sales from only 623 units it alone contributes about 11% of total revenue, which means it is a premium-priced product. In contrast, Camembert Pierrot has the highest quantity (1,577 units) among the listed products but much lower revenue, showing a high-volume, lower-price pattern.

7. Page 3 – Sales Dashboard (Interactive)

The third page, titled "Northwind Sales Dashboard", combines the key visuals from the earlier pages into one interactive screen. It repeats the KPI cards and the monthly, category and employee charts, and adds slicers so that users can filter the whole page.

Element	Purpose
KPI cards	Average Freight, Number of Customers, Total Orders, Total Freight, Total Quantity and Total Sales are shown at the top for instant reference.
Company Name slicer	Drop-down list that filters every visual to one or more customer companies.
Order Date slicer	Date range slider that restricts the analysis to a chosen period (the slider on the dashboard is set from 04-07-2013 to 04-05-2015).
Cross-filtering	Clicking an employee bar or a category bar filters all other charts and cards on the page.
Tool-tip page	Hovering over a visual displays extra detail cards (see Section 8).

Table 6: Interactive elements on the Sales Dashboard page

This page turns the report from a static summary into an exploration tool. A manager can, for example, select a single customer company, narrow the date range to one year, and immediately see that customer's orders, quantity, freight and sales without building a new report.

8. Page 4 – Tool Tip

The fourth page is a custom tool-tip page. Instead of a default pop-up showing one number, hovering over a visual displays a small card with three measures for the item under the cursor. In the dashboard screenshot, hovering over the top employee bar (Margaret Peacock) shows:

Tool-tip measure	Value shown	Interpretation
Total Orders	151	Orders handled by the employee
Total Quantity	9K	Units sold through those orders
Total Sales	225 K (approx.)	Sales value generated by the employee

Table 7: Example tool-tip values for the top employee

Tool tips add depth without cluttering the main page: the viewer sees orders, quantity and sales together for any bar or point simply by hovering over it.

9. Key Findings

- **Overall performance:** Northwind generated $1.27 million in sales from 809 orders placed by 91 customers, selling 51,317 units.
- **Customer value:** Each customer placed about 9 orders on average and generated roughly $13,900 in sales.

- **Order value:** The average order is worth about $1,565 and contains about 63 units.
- **Seasonality:** Orders are highest in the first quarter and lowest in May–July.
- **Category strength:** Beverages and Dairy Products lead sales, together around 40% of revenue.
- **Product concentration:** Côte de Blaye alone contributes about 11% of total sales.
- **Employee performance:** Margaret Peacock handles about 19% of all orders; performance varies widely across the team.
- **Geographic concentration:** The USA, France and Germany account for about 38% of customers.
- **Shipping cost:** Freight is about 5% of sales, with an average of 79.05 per order.

10. Recommendations

- Plan stock and staffing around the seasonal dip in May–July, and run promotions in those months to smooth demand.
- Protect and promote high-value products such as Côte de Blaye, while reducing dependence on a single product for a large share of revenue.
- Grow the smaller markets (for example Ireland, Norway, Poland and Switzerland) through targeted marketing, since the customer base is concentrated in a few countries.
- Share the practices of top-performing employees with the rest of the sales team and review workload allocation.
- Fix orders with a blank employee so that performance reporting is complete.
- Review shipper rates and consolidate shipments to keep freight at or below the current 5% of sales.

11. Relevance to FinTech with AI

Dashboards like this one are a foundation of modern financial technology. Real-time KPIs, interactive filtering and automated reporting are used daily in banking, payments, lending and e-commerce analytics. In the context of B.Com FinTech with AI, the same data model can be extended with AI features: forecasting of monthly sales using time-series models, customer segmentation using clustering, anomaly detection on freight costs and orders, and natural-language questions through Power BI Copilot. Learning to structure data, build measures and communicate insights visually is therefore a practical skill for careers in financial analytics and data-driven decision making.

12. Conclusion

The Power BI dashboard successfully transforms the Northwind Trade database into a clear and interactive analytical tool. The Sales Overview page summarises performance through KPIs and trend charts, the Product & Customer Analysis page reveals where customers are located and which products earn the most, the Sales Dashboard page allows users to explore the data through slicers, and the Tool Tip page provides detail on demand. Together they show that Northwind is a healthy, growing trading business with strong categories and star products, but also with seasonal weakness, geographic concentration and dependence on a few key products. Acting on the recommendations above can make sales more balanced and more profitable.

Appendix – Dashboard Screenshots

The following screenshots show the Power BI dashboard pages on which this report is based.



Figure A1: Page 1 – Sales Overview

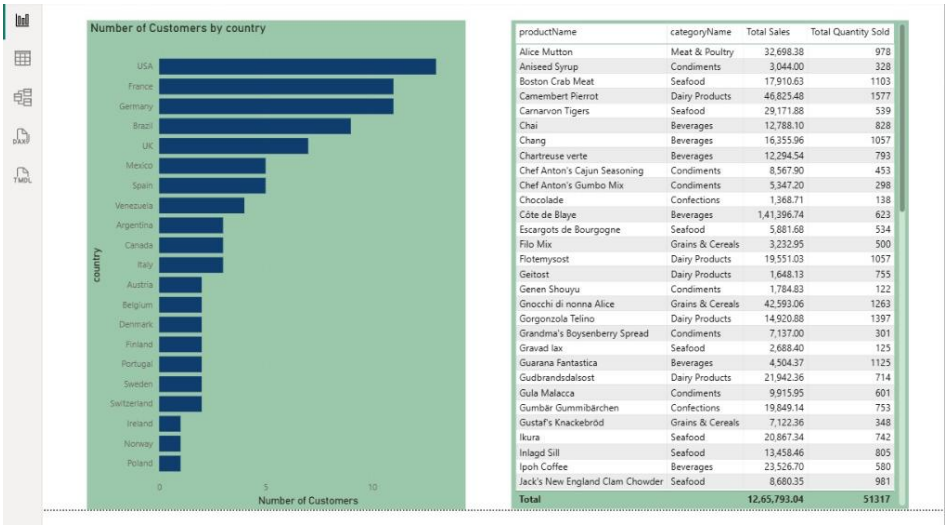


Figure A2: Page 2 – Product & Customer Analysis

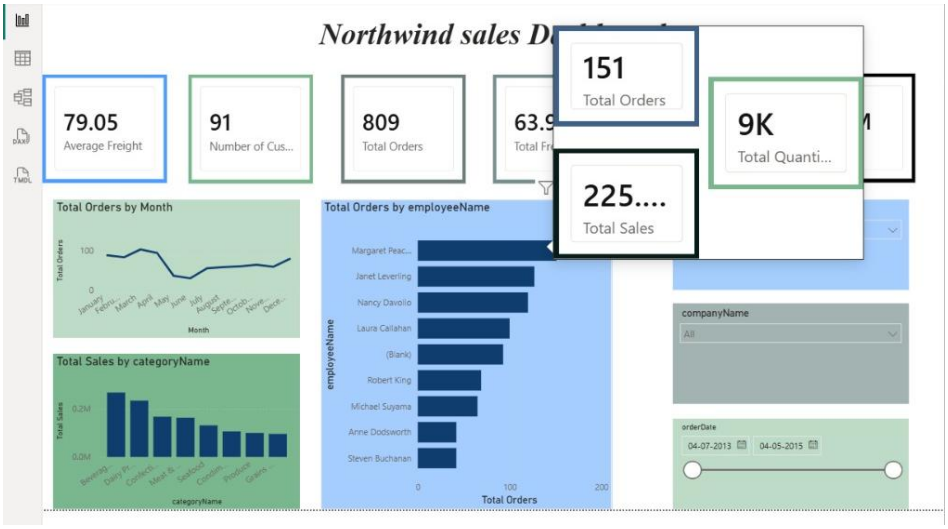


Figure A3: Page 3 – Sales Dashboard with tool-tip displayed